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BSc (Hons) in Computer Systems Engineering

IE3034 – Control Systems Engineering

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PID-Based Real-World Control System Design and Implementation

PID-Controlled Thermoelectric (Peltier) Cooling System

Design, mathematical modelling, illustrative numerical simulation and physical validation plan for a closed-loop cooling plate

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TECHNICAL STATUS: This is a fully developed cooling-oriented engineering report. Graphs and numerical values expressly labelled “illustrative simulation” are model-based calculations, not physical measurements. Actual experimental datasets and build photographs must be inserted after testing.

Abstract

This work considers the design of a compact temperature-controlled cooling surface using a TEC1-12706 thermoelectric (Peltier) module. Unlike a conventional hotplate, the cold face is thermally bonded to the target aluminium plate, while the opposite face is fixed to a heatsink with forced airflow. An Arduino Uno measures the cold-plate temperature using a DS18B20 probe and varies module drive through a properly rated logic-level MOSFET. The controller operates in the cooling direction: when measured temperature rises above the reference, the cooling demand increases. The return of warm air, a heated object or fluctuations in ambient temperature create the disturbances that feedback must reject.

The modelling starts from a lumped thermal energy balance and arrives at the first-order cooling plant $\Delta T(s)/U(s) = -K/(\tau s+1)$, together with a sensor lag and a two-state representation. For an explicitly assumed demonstration case, the values C = 160 J/°C, R = 1.0 °C/W, effective cooling capability $Q_{c,max}$ = 20 W and sensor lag τ_s = 8 s give τ = 160 s and $|K|$ = 20 °C per unit duty. These are design assumptions rather than measured plant-identification results. A constrained discrete PID with reverse-acting error, derivative-on-measurement and anti-windup is used to compare open- and closed-loop behaviour. Graphs show reference changes, a positive heat-load disturbance and separate changes to each PID gain.

The prototype design, circuit interfaces, safety considerations, proposed experiments, data sheets, reproducible MATLAB code and Arduino firmware are documented. Practical settling times, accuracy, disturbance rejection and simulation-versus-experiment errors cannot be reported responsibly until the physical system is built and logged. The report therefore marks experimental fields “To be measured” and explains how to complete them.

How to use this report

- Replace illustrative plant parameters with values identified from an open-loop cooling test; rerun the supplied MATLAB script.
- Photograph your physical assembly and capture circuit schematics from your actual build.
- Insert real logged sensor temperatures and control duty values in the experimental tables; update the comparison and conclusion.
- Verify the wiring, polarity, heat rejection and protective devices before operating the prototype.

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1. Introduction

Accurate temperature regulation is an important part of laboratory practice, electronics cooling and small-scale thermal processing. A thermoelectric cooling plate provides a manageable example because its controlled variable can be measured directly, its input can be changed electronically and its thermal dynamics are slow enough to observe in real time. However, a Peltier module does not simply “produce cold”. It pumps thermal energy from its cold face to its hot face and adds electrical dissipation to the heat which must be removed.

This distinction motivates the main engineering challenge. The target plate can only be kept below ambient if its hot side releases the extracted heat plus electrical losses effectively. A passive control circuit or a fixed PWM level cannot compensate reliably for variations in load, heatsink temperature and ambient conditions. The proposed solution uses a temperature-feedback loop with a microcontroller-based PID algorithm.

1.1 Aim

To design, model, simulate and validate a PID-based cooling system that brings a small aluminium surface below ambient temperature and holds a selected safe cooling setpoint despite moderate thermal disturbances.

1.2 Objectives

1. Define the cooling application, setpoint, manipulated input, output and disturbances.
2. Derive a lumped-parameter thermal model, transfer functions, state-space equations and feedback block diagram.
3. Simulate uncontrolled and PID-controlled behaviour and identify constraints caused by actuator saturation.
4. Explain a repeatable PID tuning method and evaluate proportional, integral and derivative actions individually.
5. Describe a buildable Arduino–sensor–power-stage prototype with a functional heatsink and safety interlocks.
6. Specify practical step-response, disturbance and uncertainty tests and compare measured records with predicted results.

1.3 Scope

The proposed device controls the temperature of a small plate below a typical indoor ambient of approximately 30 °C. Example references of 24 °C, 22 °C and 20 °C are considered for simulation. A practical operating limit is set above the risk of condensation and above the Peltier module’s capability under its actual heat load. The proposed actuator uses one current polarity and therefore provides active cooling only; warming is mainly passive when the PWM command is reduced to zero.

1.4 Report structure

Sections 2–6 establish the application, control problem and model. Sections 7–8 evaluate PID control by simulation. Section 9 describes hardware and firmware integration. Sections 10–12 provide the experimental plan, illustrative results and measurement-comparison framework. The remaining sections address interpretation, limitations, upgrades and reproducibility.

2. Description of the Real-World Application

The application is a temperature-controlled plate that can cool a small object or sample. Potential classroom demonstrations include cooling an aluminium surface to a chosen temperature, observing sensor response to a warm item and comparing a fixed-duty approach with a feedback controller. This is not a medical or refrigeration-certified device. It should not be presented as appropriate for sterile or safety-critical sample storage without additional qualification.

2.1 Why select a Peltier module?

A TEC1-12706 is a readily available solid-state thermoelectric device with two ceramic faces. When DC current flows, one face absorbs thermal energy and the other rejects energy. It contains no circulating refrigerant, but it does require a properly designed thermal path. A poor hot-side heatsink causes the hot side to heat up, decreases pumping capacity and can eventually make the intended cold plate warmer rather than colder.

For the cooling arrangement the ceramic face bonded to the aluminium plate must actually be the cold face under the selected wiring polarity; this must be checked experimentally before closing the control loop. A computer fan is used for hot-side heat rejection, not as a substitute for the thermoelectric actuator.

2.2 Proposed mechanical arrangement

[Removed placeholder illustration. Insert a real photograph of the physical cold-plate / Peltier / heatsink assembly here after prototype construction.]

The 100 mm × 100 mm × 5 mm aluminium plate is mechanically coupled to the cold face using a thin, even layer of thermal paste. The hot face must contact a sufficiently rated finned heatsink, also with thermal paste, and the fan should operate continuously whenever the TEC is powered. Compression should be evenly applied and should not fracture the ceramic Peltier faces. Insulation around the cold plate reduces unwanted heat ingress, while the sensor should be fixed away from direct warm exhaust flow.

2.3 What the controller regulates

The sensor measures the plate, not the hot-side heatsink. A rising measured temperature relative to a fixed target represents inadequate cooling. The PID therefore increases PWM duty to extract more thermal energy. Once the plate approaches the target, duty is reduced toward the amount needed to offset environmental heat gain. If the target is raised while the plate is already cold, the controller can stop cooling but cannot actively warm it in this unidirectional configuration.

3. Problem Definition

Below ambient, the surrounding room transfers heat into the plate through conduction, convection and radiation. Any object placed on the cold surface adds another heat load. Meanwhile the Peltier pumping capability depends on electrical input, the cold–hot temperature difference and hot-side rejection efficiency. The required input cannot therefore be treated as a fixed number under every load.

Control element	Cooling-specific definition	Illustrative value
Controlled variable	Aluminium plate temperature T(t)	°C; target 20–24 °C
Manipulated variable	TEC supply duty fraction u(t)	0–0.85 (0–85%)
Setpoint	User-selected plate temperature Tref	22 °C example
Disturbance	Positive heat load transferred into plate	+4 W demo
Sensor	DS18B20 digital temperature sensor	1 s readings, 12-bit
Controller	Arduino Uno executing reverse-action PID	Ts = 1 s
Actuator	MOSFET-powered thermoelectric module	nominal 12 V supply
Hot-side management	Fan and heatsink	Independent continuous operation

3.1 Why feedback is necessary

With a fixed 40% duty, the assumed model ultimately reaches 22 °C when the room is held at 30 °C and the thermal parameters remain unchanged. In practice those assumptions fail when airflow changes, a warm object is applied or the module hot side warms up. A feedback loop detects the plate-temperature error and compensates within the limited cooling authority available. Integral control is helpful for rejecting sustained heat loads, but windup must be prevented when the command is saturated.

3.2 Performance targets

Requirement	Engineering design target	Validation method
Cooling settling time	Within 5 min for 30 → 22 °C, if hardware capability permits	Step response log
Undershoot below setpoint	≤ 5% of specified 8 °C step	Minimum temperature
Steady error	Within ±0.5 °C	Last 2 min average
Heat-load deviation	≤ 2 °C for proposed +4 W load	Warm-object / calibrated heater trial
Sensor safety	Turn module off on missing sensor / unsafe hot side	Disconnect and overtemperature tests
Practical repeatability	Similar results across three repetitions	Overlay repeated runs

These are proposed specifications, not assertions that the final prototype has already met them.

4. Control-System Design

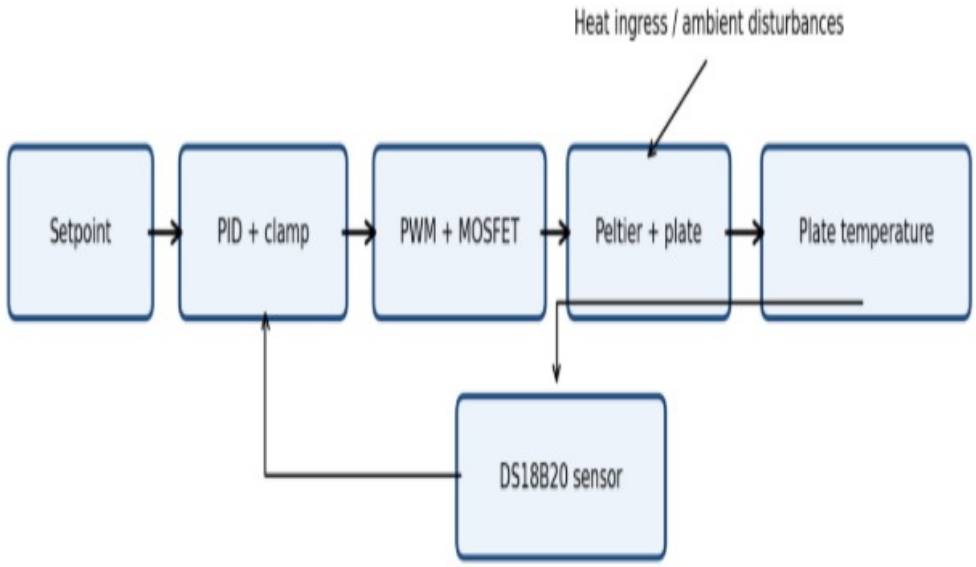


Figure 2. User-supplied closed-loop control block diagram of the Peltier cooling system.

The uploaded block diagram shows the implemented cooling-control sequence clearly: the user-set setpoint enters the PID controller with output clamping, the PWM and MOSFET stage drives the Peltier module, the Peltier-plus-plate plant produces the controlled plate temperature, and the DS18B20 sensor closes the feedback loop. Ambient heat ingress and external heat loads act as disturbances on the plant. This is a suitable high-level control-structure figure for the report because it reflects the real system architecture more naturally than an illustrative placeholder.

4.1 Major engineering choices

- One-second temperature sampling matches the DS18B20 conversion delay and is substantially faster than the assumed 160-second thermal time constant.
- The power switch should be a suitable low-RDS(on), logic-level N-MOSFET with appropriate heatsinking and a safe gate-drive arrangement. High TEC currents require a supply, wiring and protection sized for actual conditions.
- A fan failure or excessively hot heatsink can remove almost all effective cooling capacity. Monitor the hot-side temperature independently where possible.
- The control error has the opposite sign to the heater case. Reusing hotplate firmware without correcting the sign would create positive feedback.
- The controller uses derivative-on-measurement, a filtered derivative estimate and conditional integration to limit setpoint kick, noise and integrator windup.

4.2 Operating states

State	Entry condition	Control response
Idle	Valid sensor; no cooling requested	TEC off; fan policy as appropriate
Active cooling	Plate above target and all checks pass	TEC PWM regulated by PID
At setpoint	Measured temperature close to target	Maintain necessary duty
Sensor fault	DS18B20 missing / implausible reading	TEC immediately off
Hot-side thermal fault	Heatsink limit exceeded	TEC immediately off; fan remains on
Power recovery	After restart	Reset integrator; validate sensors first

5. Mathematical Modelling

5.1 Modelling assumptions

7. The aluminium plate has one lumped temperature because of its high conductivity and modest thickness.
8. An equivalent thermal resistance R aggregates ambient-to-plate heat inflow near one operating condition.
9. The net heat-pumping action of the Peltier is locally proportional to duty: $Qc \approx Qc,max\ u$. This approximation is only valid near a chosen hot-side temperature and temperature difference.
10. Electrical and thermal parameters are constant over a single short test; changes are treated as modelling errors.
11. The sensor and attachment introduce a first-order measurement lag.
12. Forced-air heat rejection is available while cooling operates; condensation and icing are excluded from the nominal model.

5.2 Variables and assumed numerical values

Symbol	Meaning	Nominal assumption
T	True plate temperature	°C
Tm	Measured plate temperature	°C
Ta	Ambient temperature	30 °C
u	TEC PWM fraction	0–0.85
C	Effective thermal capacitance	160 J/°C
R	Plate to ambient thermal	1.0 °C/W

	resistance	
Qc,max	Effective heat pumped at unit duty	20 W
τ = RC	Dominant plate time constant	160 s
K = R Qc,max	Absolute plant gain	20 °C per duty
τm	Temperature sensor lag	8 s
Qd	Extra heat load into plate	W, positive warms

The plate itself has approximate mass ρV = 2700 × 0.1 × 0.1 × 0.005 = 0.135 kg. At cp ≈ 900 J/(kg·°C), its thermal capacitance is about 121.5 J/°C. An illustrative combined value of 160 J/°C allows for thermal paste, fasteners and sensor contact. It is not a measurement of an assembled device.

$$C_{plate} = \rho V c_p \approx 121.5 \text{ J/°C}; \quad C_{effective} = 160 \text{ J/°C (assumed)}$$

5.3 Energy-balance equation

Rate of thermal energy stored in the plate equals heat entering from the room plus the imposed heat-load disturbance, minus the heat pumped away by the TEC. For a positive cooling input, the plant gain from duty to plate temperature must be negative:

$$C\,dT/dt = (T_a - T)/R + Q_d - Q_{c,max}\,u$$

Using temperature deviation θ = T − Ta with approximately constant ambient temperature:

$$\tau\,d\theta/dt + \theta = -K\,u + R\,Q_d, \quad \tau = RC, \quad K = R\,Q_{c,max}$$

The temperature sensor is approximated by:

$$\tau_m\,dT_m/dt + T_m = T$$

5.4 Limits of a simple model

Real TEC heat pumping is nonlinear: it depends on face temperatures, applied current, Joule heating, Seebeck transport and back conduction. Because hot-side thermal resistance can vary substantially, measured plant parameters should be identified experimentally at the operating point. The first-order expression is used here for transparent control design and initial tuning, not as a substitute for an experimental characterization.

6. Transfer Function / State-Space Model

6.1 Transfer-function derivation

Taking Laplace transforms of the deviation equations with zero initial deviations gives two independent paths: duty input (cooling) and applied heat disturbance (warming).

$$G(s) = \Theta(s)/U(s) = -K/(\tau s + 1) = -20/(160s + 1)$$

$$G_d(s) = \Theta(s)/Q_d(s) = R/(\tau s + 1) = 1/(160s + 1)$$

$$H(s) = T_m(s)/T(s) = 1/(\tau_m s + 1) = 1/(8s + 1)$$

$$G(s)H(s) = -20 / [(160s+1)(8s+1)]$$

The negative sign is physically meaningful: more cooling produces a lower plate temperature. The characteristic open-loop poles are $-1/160 = -0.00625\ s^{-1}$ and $-1/8 = -0.125\ s^{-1}$. The uncontrolled thermal model is stable but responds relatively slowly.

6.2 State-space representation

Choose state $x = [\theta, \theta m]^T$ with temperature deviations relative to ambient and measured deviations. Use the two inputs $[u, Qd]^T$. The measured output is $y = \theta m$:

$$\dot{x} = A\,x + B\,[u\ \ Qd]^T; \quad y = [0\ \ 1]\,x$$

$$A = [[-0.00625, 0], [0.125, -0.125]]$$

$$B = [[-0.125, 0.00625], [0, 0]]; \quad C_out = [0, 1]; \quad D = [0, 0]$$

The duty-input controllability matrix has full rank because the plate temperature drives the sensor state, and both states affect the measurement after the sensor dynamics. This confirms that the two-state nominal linear system can be observed and influenced through the available sensor and cooling input. Control sign must still be handled consistently.

6.3 Steady-state example

With 30 °C ambient, no extra heat disturbance and the nominal assumption K = 20 °C/duty, fixed u = 0.40 gives Tss = 30 − 20 × 0.40 = 22 °C. A +4 W sustained heat disturbance shifts the temperature upward by R Qd = 4 °C unless the controller increases duty. In this simple linear model, the duty must rise by Δu = 4/20 = 0.20 (20 percentage points), provided the actuator is not saturated.

7. MATLAB/Simulink Simulation

All figures in this section were generated by numerical integration of the explicitly stated nominal model. They are illustrative simulated predictions, not instrument data collected from a built unit. The simulation samples PID calculations at 1.0 s, integrates thermal states at 0.1 s for presentation, and constrains TEC duty to 0–85%. A corresponding MATLAB workflow is provided in Appendix B.

7.1 Simulation set-up

Block / setting	Cooling model configuration
Setpoint source	22 °C fixed or 24 → 20 → 24 °C sequence
Controller	Reverse-acting PID: e = Tmeas − Tref
Plant	−20/(160s+1)
Sensor	1/(8s+1)
Disturbance	Positive heat load through +1/(160s+1)
Saturation	0 ≤ PWM duty ≤ 0.85
Sample interval	1 second PID, 0.1 second thermal integration
Initial condition	30 °C plate and sensor

7.2 Before applying feedback

For fixed 40% duty, the nominal plate approaches 22 °C with a dominant 160-second thermal lag. Without feedback the apparent equilibrium depends on thermal resistance and net pumping capability; a sustained external heat input shifts the final temperature. The open-loop trace illustrates why a calculated fixed duty alone is not a robust regulator.

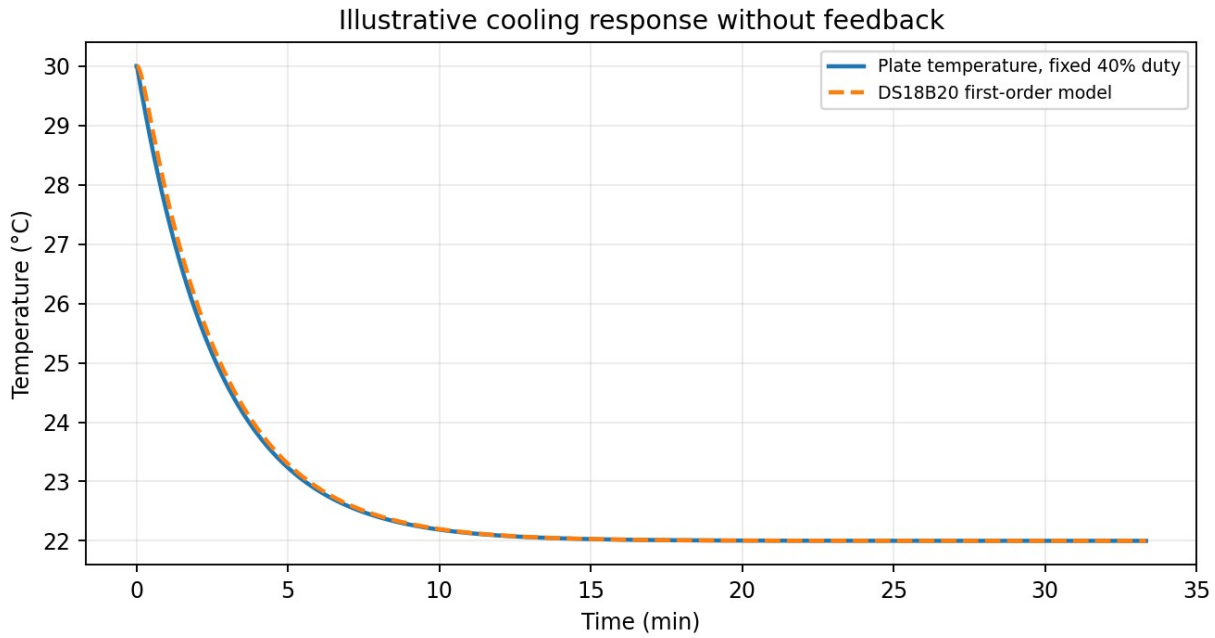


Figure 3. Illustrative fixed-duty cooling transient from 30 °C toward 22 °C.

7.3 Applying PID feedback

The model initially demands substantial cooling while the plate is far above target. As the measured temperature approaches the setpoint, the output falls toward the steady cooling requirement. Saturation prevents demanding more than the selected safe command ceiling. The graph illustrates a controlled transient under the assumed thermal model, but actual hot-side behavior may be significantly different.

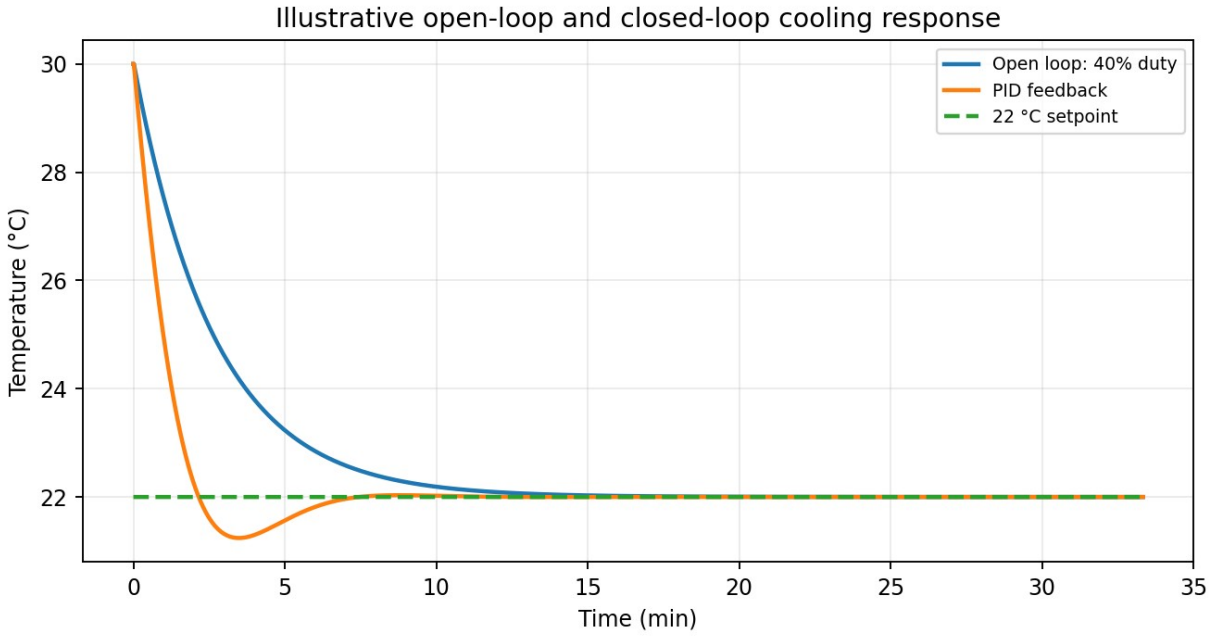


Figure 4. Illustrative comparison of fixed-duty open loop and feedback PID cooling.

7.4 Setpoint changes and disturbances

The setpoint is first 24 °C, then reduced to 20 °C at 20 minutes, and raised again to 24 °C at 40 minutes. A +4 W heat-load input is added at 50 minutes and removed after approximately 61.7 minutes. The downward reference step calls for more cooling, whereas the upward reference requires reduced duty and may temporarily rely on passive warming. The heat disturbance is positive because it makes the cold surface warmer.

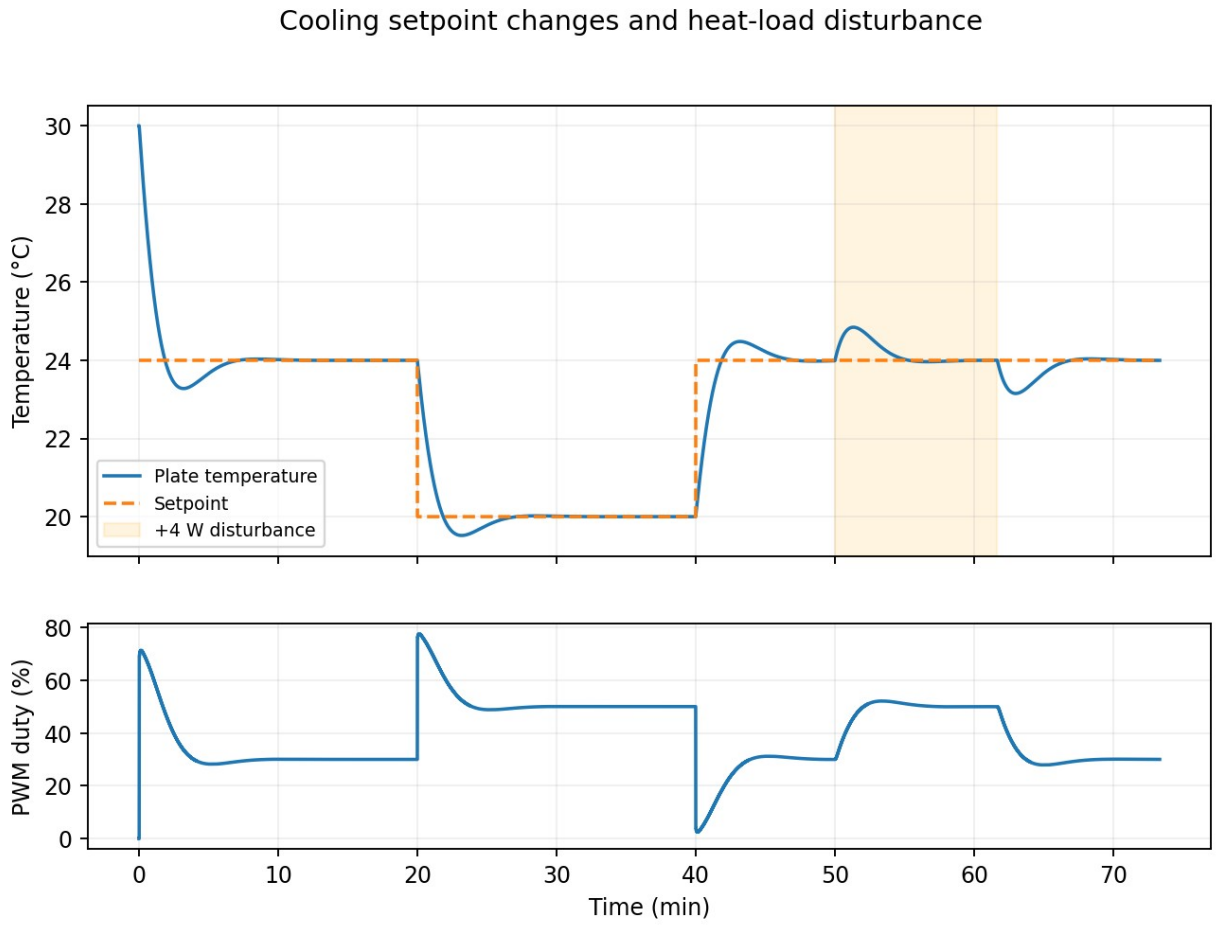


Figure 5. Illustrative plate and PWM response to reference changes and a +4 W heat-load disturbance.

7.5 Simulation evidence supplied for the cooling prototype

The following four figures are the MATLAB/Simulink outputs supplied by the project group. They are presented separately from the illustrative nominal-model results in Figures 3–10, since the corresponding Simulink parameter file and raw timeseries were not provided. Therefore, all numerical readings below are approximate visual estimates from the figures, not independently verified measured data.

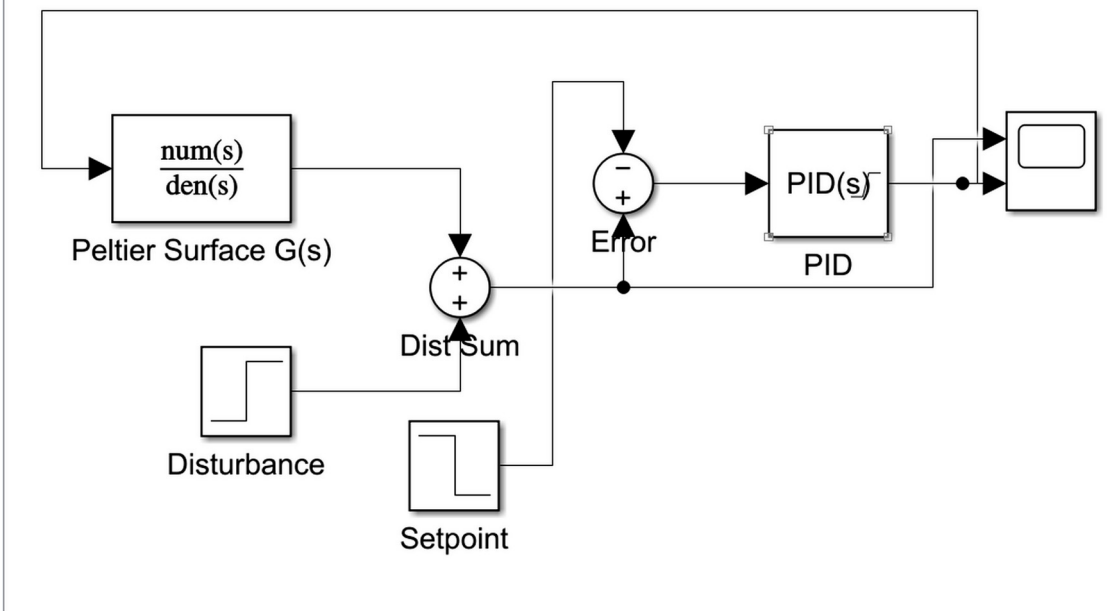


Figure 11. User-supplied Simulink model of the Peltier surface-temperature control loop.

This Simulink model contains the core elements of the software study: a Peltier surface transfer-function block G(s), a disturbance step input, a setpoint step input, a summing junction that forms the control error, a PID block and a scope to observe the resulting response. In the final report, this figure should be described as the actual simulation model used by the group. During viva, explain the signal flow clearly: the setpoint and measured temperature generate the error, the PID computes the control action, the plant block represents the cooling dynamics, and the disturbance input is used to test disturbance rejection.

Figure 12. Submitted PID temperature trajectories for 5, 10, 15 and 20 °C setpoints; disturbance begins at approximately t = 1200 s.

All cases begin around 25 °C. The 15 and 20 °C responses initially fall below their targets before recovering. The 10 °C case reaches approximately 10 °C by the disturbance time, but it then jumps to around 13 °C and remains there. The 5 °C case also approaches roughly 10 °C and stays there, indicating that 5 °C is not achievable with the parameters and control limits represented in this simulation. The 15 °C and 20 °C traces recover toward their reference temperatures after the disturbance.

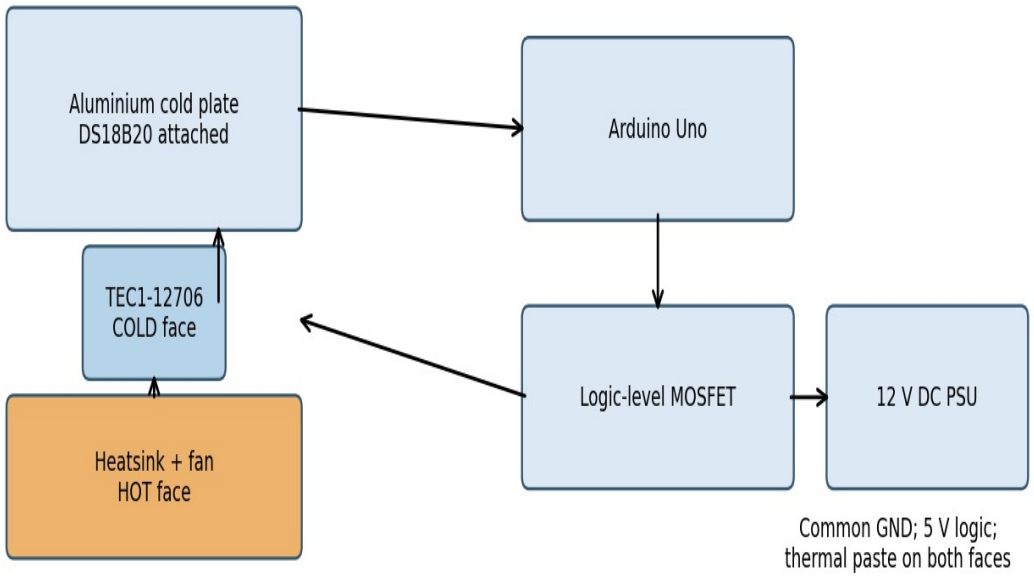


Figure 13. Submitted comparison of PI and PID surface-temperature responses at four cooling setpoints.

For 5 and 10 °C, the PI and PID traces almost completely overlap and fail to demonstrate consistent regulation at the 5 °C target. At 15 °C, the PI trace has a somewhat deeper initial undershoot than the PID trace; both subsequently recover. At 20 °C, PI shows somewhat larger undershoot and a slightly stronger corrective excursion than PID. These visible trends suggest a modest damping benefit from derivative action in attainable cases, not a universal improvement under actuator saturation.

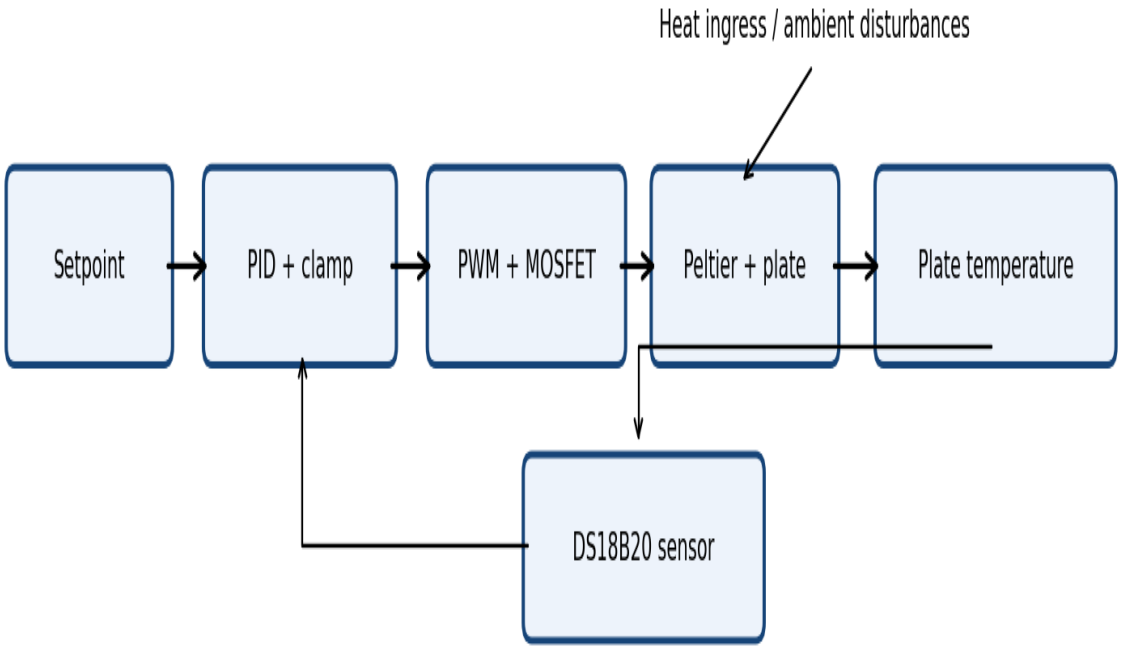


Figure 14. Submitted PI and PID PWM duty-cycle responses (0–1 duty fraction).

Both controllers demand near-unity duty cycle throughout the 5 and 10 °C runs. Consequently, these two runs primarily test the actuator limit rather than the unconstrained controller tuning. For 15 °C, the initial effort reaches nearly 1 before converging to approximately 0.66; after the disturbance it rises close to 1 and settles near 0.86. For 20 °C, the duty is about 0.33 before the disturbance and approximately 0.53 afterward. These are graph estimates. Actual power/current and thermal capacity must be established experimentally.

7.6 Interpretation and verification of supplied Simulink outputs

Reference	Before t = 1200 s	After disturbance	Interpretation
5 °C	Approximately 10 °C plateau	Approximately 13 °C plateau	Unreachable target; 100% duty
10 °C	Approximately 10 °C	Approximately 13 °C plateau	Disturbance cannot be rejected under saturation
15 °C	Settles near 15 °C	Recovers toward 15 °C	Attainable under plotted conditions
20 °C	Settles near 20 °C	Recovers toward 20 °C	Attainable under plotted conditions

The roughly instantaneous 3 °C upward jump at t ≈ 1200 s needs explanation. A real aluminium plate with finite thermal capacitance cannot physically change by several degrees instantaneously when a heat-load disturbance is applied. The plotted signal may include an additive temperature offset

introduced downstream of the thermal plant, or may represent a measurement disturbance. To model a physical thermal load, add disturbance power (watts) within the energy-balance dynamics, upstream of the temperature state; a true thermal response should then be continuous. Document the exact Step amplitude and units in the Simulink model.

It is also important to check the output saturation configuration. The previously drafted example used 0–0.8 duty, whereas the supplied duty plots reach 1.0. These represent different controller implementations and cannot be directly compared unless the limits, Kp, Ki, Kd, ambient temperature and Peltier parameters are made consistent. Figures 11–14 are therefore treated as the group's submitted simulation evidence, while the earlier nominal graphics are illustrative modelling examples.

8. PID Controller Design

8.1 Reverse-acting control law

The controller must increase duty when measured temperature is too warm. Therefore use e[k] = Tm[k] − Tref[k], in contrast with a heater controller. The parallel control law is:

$$u^*(t) = Kp\, e(t) + Ki \int e(t)\, dt + Kd\, (dTm/dt)$$

Derivative-on-measurement provides a rate-dependent contribution while avoiding an abrupt derivative kick when the target changes. Positive dTm/dt means the plate is warming and can justify more cooling. The resulting raw output is limited physically to the selected PWM range.

8.2 Example gains and tuning workflow

A cautious starting set of illustrative discrete gains is Kp = 0.115 duty/°C, Ki = 0.0015 duty/(°C·s) and Kd = 0.55 duty·s/°C, with a filtered derivative. These values are demonstration gains for the assumed plant, not measured or optimally tuned gains for a TEC1-12706 prototype. A sensible field procedure is to identify the open-loop response first, begin with low proportional action, add integration gradually to remove offset, and add limited derivative action only if it improves damping.

Term	Illustrative gain	Effect
Kp	0.115 (duty/°C)	Immediate correction to excess temperature
Ki	0.0015 (duty/(°C·s))	Compensates constant thermal load
Kd	0.55 (duty·s/°C)	Anticipates warming/cooling rate
PWM saturation	0 to 0.85	Prevents impossible demand
Derivative filter	First-order low-pass on slope	Reduces sensor-quantization sensitivity

8.3 Comparison with other controller forms

A proportional controller usually leaves a steady offset under constant load because duty is proportional to error. PI can remove the steady offset if required steady duty is inside the allowable range. Derivative action can damp rapid adjustments, but overlarge derivative gain amplifies measurement noise. Strong integral gains often cause excess undershoot after an extended period of actuator saturation.

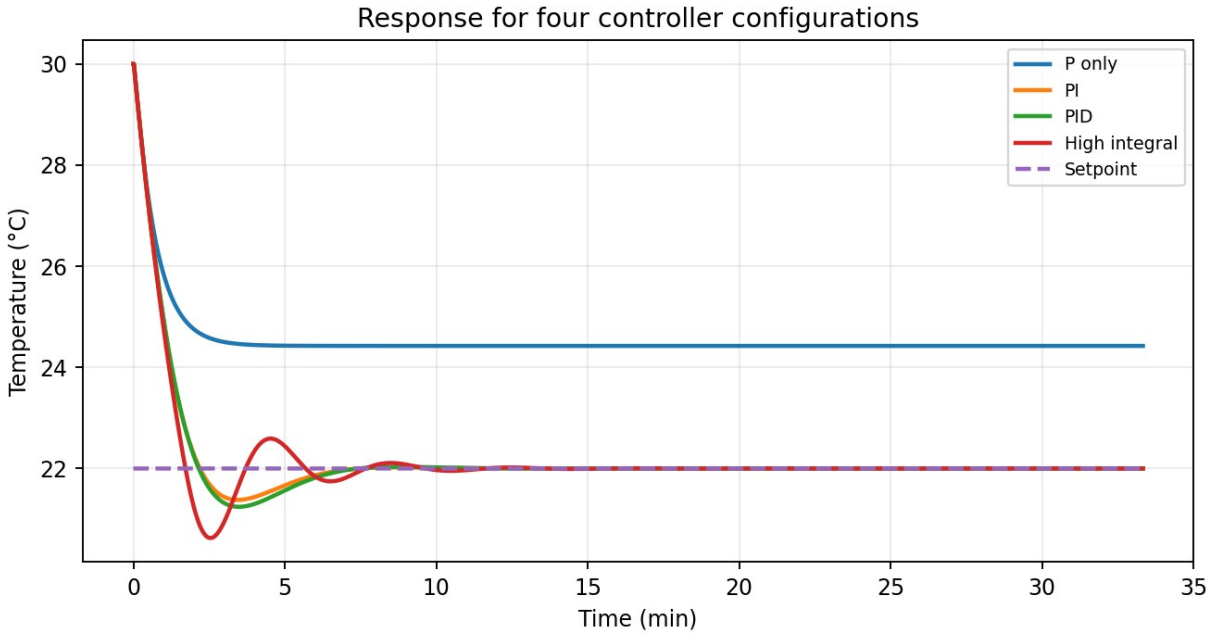


Figure 6. Illustrative P, PI, PID and aggressive-integral cooling responses.

8.4 Effect of varying Kp, Ki and Kd

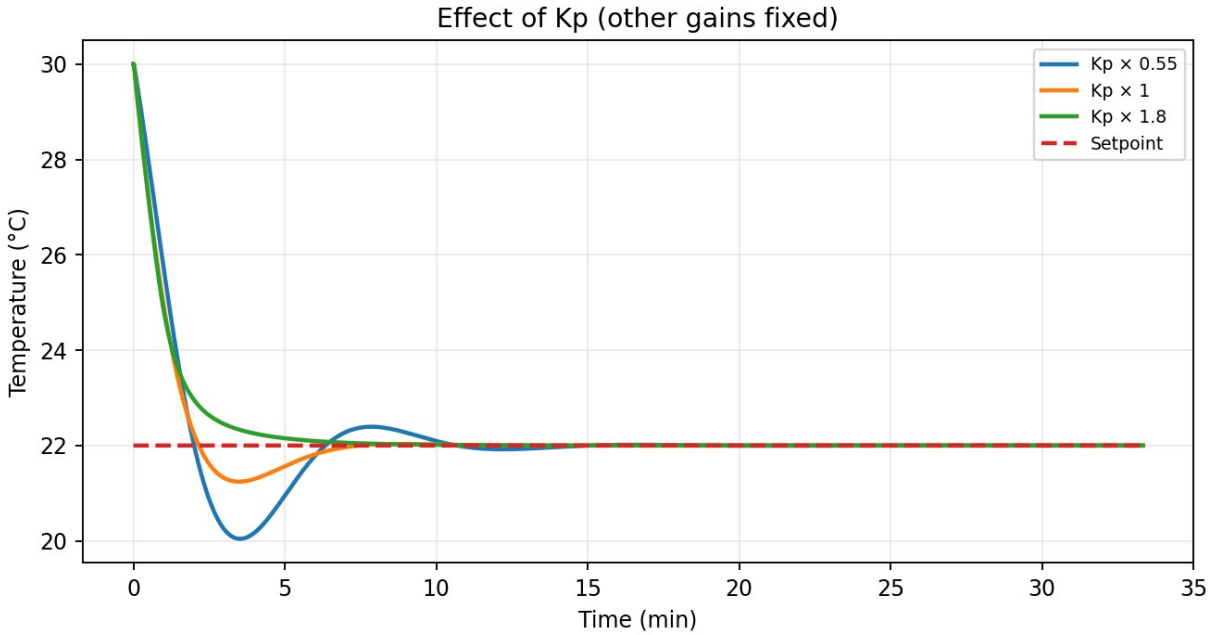


Figure 7. Illustrative sensitivity to proportional gain.

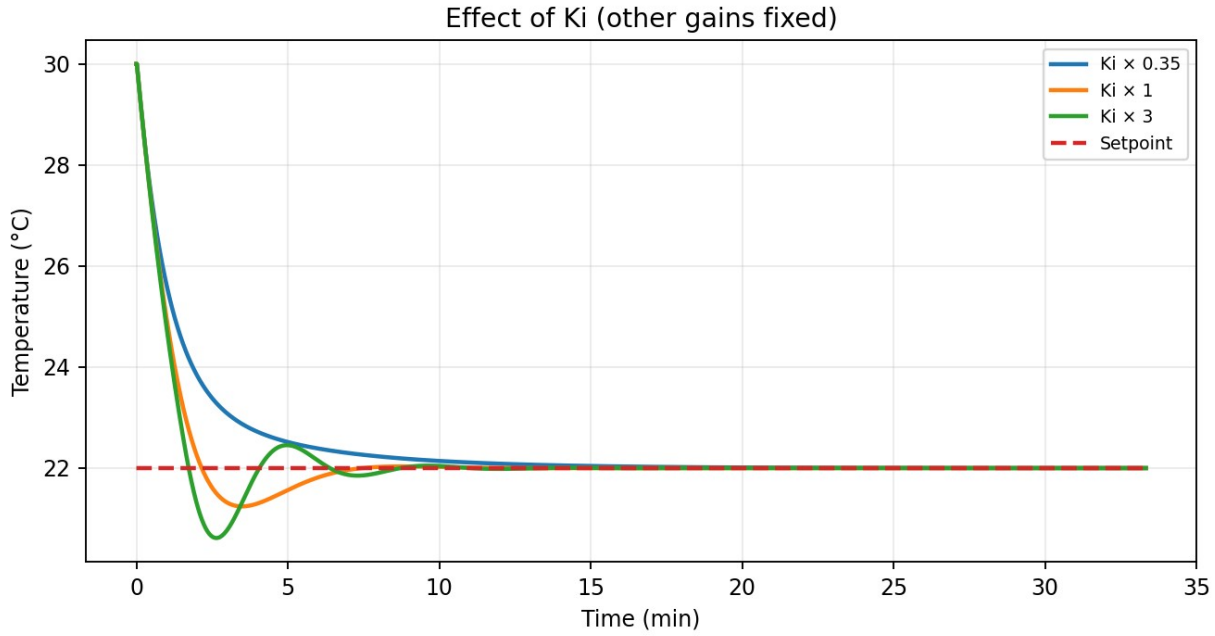


Figure 8. Illustrative sensitivity to integral gain.

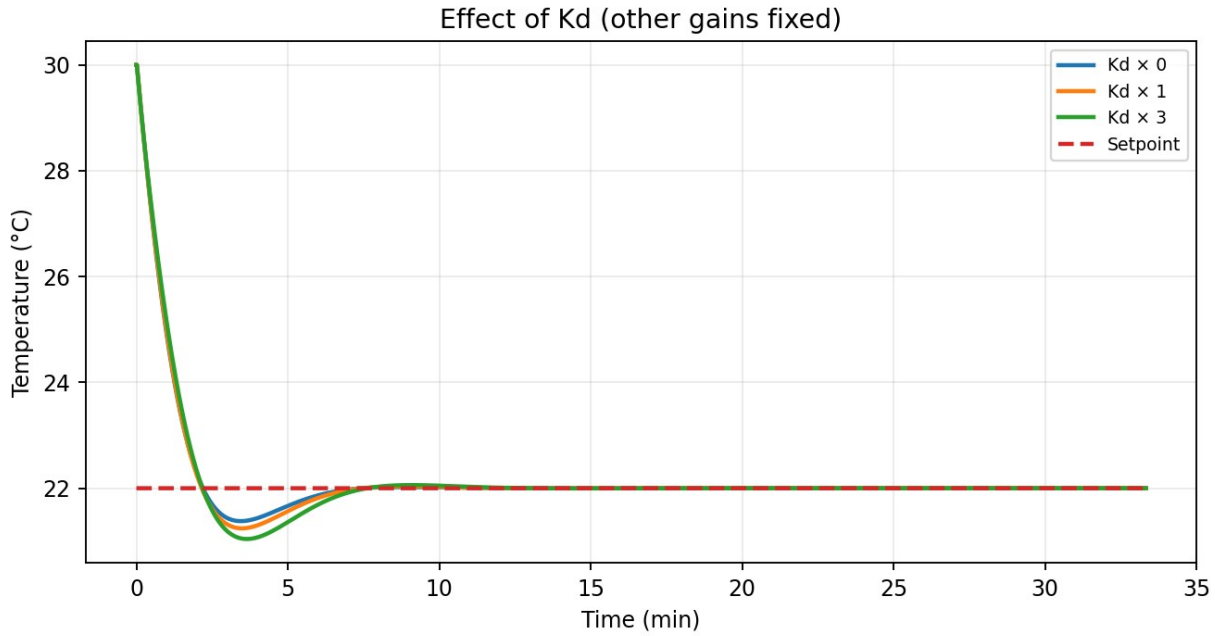


Figure 9. Illustrative sensitivity to derivative gain.

Parameter increased	Likely benefit	Potential drawback
Kp	Faster response to excess temperature	Noise, output saturation, undershoot
Ki	Removal of sustained offset	Windup and oscillatory correction
Kd	Rate damping and reduced transient swing	Sensor-noise amplification, sluggishness

8.5 Stability and implementation considerations

A nominal frequency-response calculation can be applied to the linearized plant and controller. The phase plot shown is exploratory because the practical loop includes saturation, PWM switching, sampled sensing, hot-side thermal dynamics and a derivative filter. Margins derived from the simplified transfer function cannot replace direct stable-operation verification over the expected load and temperature range.

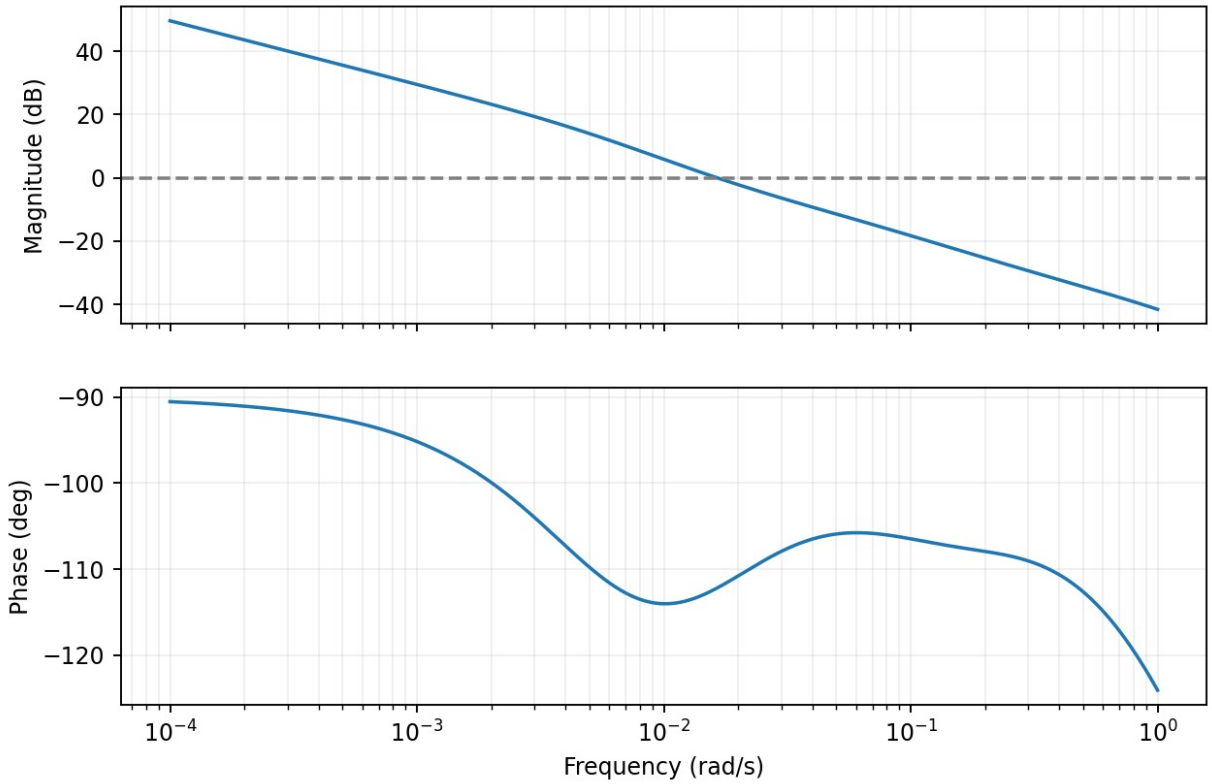


Figure 10. Nominal linear loop-frequency response (illustrative; nonlinearities excluded).

Anti-windup stops the integrator from accumulating in the same direction when the commanded PWM is already at a limit. The program resets the integrator following an invalid sensor or thermal fault, turns the TEC off, and requires stable conditions before active cooling resumes.

9. Hardware Design and Implementation

9.1 Bill of materials

Item	Example specification / purpose	Verification point
Arduino Uno	5 V microcontroller for PID and PWM	Compatible pins and ground
TEC1-12706	Thermoelectric actuator	Correct cold-face polarity
DS18B20	Waterproof 1-Wire probe	12-bit read; 4.7 kΩ pull-up

N-MOSFET	Logic-level, low-RDS(on), rated for DC switching	5 V gate drive; thermal limits
Power supply	12 V DC, enough current for TEC and fan	Measured current and fuse
Hot-side heatsink	Large finned heatsink plus fan	Surface remains within safety limits
Cold plate	100 × 100 × 5 mm aluminium	Flat, insulated, secure mounting
Thermal paste	Thin thermal interface layers	Both TEC faces
Power protection	Inline fuse, bulk capacitor, connectors	Wire gauge and polarity
Optional hot-side sensor	Second thermometer / NTC	Overheat fault detection

9.2 Power and sensor interfaces

The DS18B20 data line connects to a designated Arduino digital input (example D2) with a 4.7 kΩ pull-up to 5 V; sensor ground and Arduino ground are common. The MOSFET gate is driven through a modest series resistor (for example 100 Ω) with a gate-to-ground pull-down (for example 10 kΩ). TEC positive connects to the fused 12 V supply and TEC negative to the MOSFET drain in a low-side arrangement; MOSFET source returns to supply ground. The gate wiring and MOSFET thermal design must be verified for the actual component and PWM waveform.

Unlike an inductive relay coil, a bare TEC module is predominantly resistive. A conventional flyback diode directly across the TEC is not inherently required for the module itself; protection for fans, relays and other inductive loads must be evaluated separately. Keep switching-current wiring short and route sensor wiring away from high-current return paths. Where a MOSFET switches the TEC, choose PWM frequency and power-stage design suitable for thermoelectric drive; very high ripple currents may reduce efficiency.

9.3 Connection table (functional, verify against actual schematic)

Source pin	Destination	Role
Arduino 5 V	DS18B20 VDD	Sensor supply
Arduino GND	DS18B20 GND / PSU negative	Common reference
Arduino D2	DS18B20 DATA	1-Wire input
4.7 kΩ	5 V to DATA	1-Wire pull-up
Arduino D9	Gate resistor → MOSFET gate	PWM output
10 kΩ	MOSFET gate → GND	Prevents floating gate
MOSFET source	PSU negative	Power return
MOSFET drain	TEC negative	Switched return

Fused +12 V	TEC positive	Module supply
Fan supply	Suitable rated output	Continuous heatsink cooling

This table is a design guide, not proof of an electrically verified production schematic. The connection orientation and the physical hot/cold TEC face must be checked on the assembled prototype.

9.4 Mechanical assembly and cooling precautions

13. Confirm the heatsink/fan assembly is suitable for the combined pumped heat and electrical input.
14. Apply thermal paste to both contact interfaces and clamp with even pressure.
15. Attach the temperature probe firmly to the cold plate and insulate its rear/edges if practical.
16. Use a fan supply and airflow path independent of the TEC PWM command.
17. Test the selected TEC polarity briefly while measuring both faces; do not run without heatsinking.
18. Implement a hot-side temperature cutoff and consider condensation on surfaces below the dew point.

9.5 Embedded firmware

Appendix A contains a detailed Arduino sketch with a one-second sampling interval, serial setpoint input, temperature validity checks, integral clamping, derivative filtering, bounded PWM output and optional hot-side limit. It assumes a correctly sized MOSFET power stage; the firmware cannot by itself protect an inadequately rated heatsink or power supply.

10. Experimental Methodology

The experiments below are procedures to run on the completed prototype. They are not claimed as already completed. For valid simulation-to-hardware comparison, log actual ambient temperature, heatsink temperature, plate temperature, sensor value and PWM duty, and preserve the raw CSV file.

10.1 Equipment and conditions

- Prototype with both thermal interfaces secured, guarded wiring and fan working.
- USB-connected laptop for timestamped Arduino Serial logging at 1 Hz.
- Independent reference thermometer for sensor validation.
- A controllable warm object or known small heater to introduce a repeatable disturbance.
- Room thermometer for Ta, and a second temperature probe on the heatsink.

10.2 Experiment 1: fixed-duty step and identification

19. Allow the device to reach ambient temperature with the TEC off.
20. Record Ta, plate temperature and hot-side temperature before applying cooling.
21. Apply a moderate constant duty (example 40%) with the fan continuously powered.
22. Record temperature at 1 s intervals until sufficiently close to the final value; abort if hot side overheats.
23. Calculate $K = |Tss - Ta|/u$ and estimate τ from the time to achieve approximately 63.2% of the total temperature change.

24. Repeat at several safe duty values to assess linearity and select the operating region for feedback tuning.

10.3 Experiment 2: closed-loop reference steps

25. Start from ambient and command 24 °C. Log the transient and steady portion.
26. After temperature stabilizes, command 22 °C, followed by 20 °C only if safe and achievable.
27. Raise the setpoint back to 24 °C to document passive-warming asymmetry.
28. Repeat each relevant step three times and calculate settling time, minimum temperature, steady-state error and required duty.

10.4 Experiment 3: warm-object disturbance rejection

Once the plate has stabilized at the test setpoint, introduce a controlled repeatable warm load at a recorded time. Log the maximum positive deviation from target, recovery duration and change in PWM. Remove the disturbance at a recorded time. If the applied heat is not measured, describe the object and conditions rather than presenting it as a known wattage.

10.5 Experiment 4: variation of Kp, Ki and Kd

Change one controller gain at a time, preserving other values and the test conditions. Use moderate changes first; large integral gains may cause severe undershoot or prolonged saturation. Include at least one P-only and one PI comparison, and document the controller output as well as temperature.

10.6 Metrics and calculations

$$e_{ss} = T_{ref} - \text{mean}(T_{measured\ in\ final\ stable\ window})$$

$$\text{Cooling undershoot (\%)} = \max(0, T_{ref} - T_{min})/|T_{initial} - T_{ref}| \times 100\%$$

$$\text{Settling time} = \text{first time after which } |T - T_{ref}| \leq 0.02 \times |\text{step}| \text{ continuously}$$

$$\text{Maximum heat-load deviation} = \max(T - T_{ref}) \text{ during applied disturbance}$$

Rise time for cooling can be defined as the time to complete 10–90% of the downward temperature transition. The tolerance convention, sensor noise and logging interval must be specified when reporting these values.

10.7 Safety precautions

Never energize an unheatsinked TEC module. Verify the power supply current rating, fuse, polarity, wire temperature and MOSFET case temperature. Protect the controller from liquid condensation and do not run the cold plate below the ambient dew point without precautions. Interrupt TEC power if the temperature sensor fails, hot side gets too hot, fan stops or condensation threatens electrical equipment.

11. Results

This section separates numerical predictions produced from the assumed model from results that require laboratory measurements. No genuine prototype measurements were supplied. The physical-results fields deliberately remain unfilled rather than copying heating data from the original document or labelling simulated values as experimental evidence.

11.1 Nominal system-identification prediction

Quantity	Nominal model prediction	Actual prototype
Initial plate temperature	30 °C	To be measured
Fixed input	40% PWM	To be recorded
Steady cooling temperature	22 °C	To be measured
Plant gain magnitude	20 °C per duty	To be identified
Time constant	160 s	To be identified
Sensor lag	8 s	To be identified

11.2 Nominal closed-loop step prediction

Metric, 30 → 22 °C	Illustrative PID simulation	Experimental value
90% of cooling transition	102.7 s	To be measured
2% settling time	367.3 s	To be measured
Cooling undershoot	9.48%	To be measured
Final plate temperature	22.00 °C	To be measured
Final absolute target error	0.000 °C	To be measured

Values above follow from the assumptions C = 160 J/°C, R = 1.0 °C/W, Qc,max = 20 W and the candidate PID gains. If actual effective cooling decreases because the hot side is warmer than expected, laboratory transients can be slower and equilibrium temperatures can differ substantially.

11.3 Disturbance rejection prediction

The analytical nominal open-loop steady-state effect of a +4 W heat input is +4 °C (R × Qd). The controller should respond by increasing duty by about 0.20 at equilibrium if there is sufficient output headroom. The plotted trajectory in Figure 5 is the numerical closed-loop example; its detailed peak and recovery characteristics should be recalculated after the model is identified.

11.4 PID gain variations

Figures 6–9 illustrate the differences between P, PI, PID and aggressive gain settings. The specific trends are plant- and saturation-dependent and should not be treated as universal evidence that one gain combination is optimal for the physical prototype. A justified choice requires repeating the practical tests and evaluating errors, oscillation, controller effort and hot-side temperature.

11.5 Sensor accuracy

The commonly specified DS18B20 accuracy is approximately ±0.5 °C over much of its normal range, and the 12-bit quantization step is 0.0625 °C. Quantization is not the same as absolute accuracy. A well-placed reference thermometer should be used to check probe offset at several stable temperatures, accounting for contact error and different probe time constants.

11.6 Results extracted from supplied simulation plots

The group's plots allow qualitative conclusions but do not supply the original data arrays. Therefore, exact rise time, 2% settling time, maximum undershoot and integral error cannot be calculated reliably from the images alone. The table below reports only visually supported outcomes.

Setpoint	Steady-state outcome before disturbance	Observed PID control effort
5 °C	≈ 10 °C; about +5 °C tracking error	≈ 1.0, continuously saturated
10 °C	≈ 10 °C; small pre-disturbance error	≈ 1.0, continuously saturated
15 °C	≈ 15 °C; near-zero steady-state error	≈ 0.66 before / 0.86 after
20 °C	≈ 20 °C; near-zero steady-state error	≈ 0.33 before / 0.53 after

The persistent temperature plateau near 10 °C for the 5 °C case is a practical control insight: integral action cannot force a saturated Peltier module beyond the available cooling capacity. Anti-windup should therefore be used to prevent integrator accumulation and undesirable recovery when the reference is later changed.

12. Results Comparison – Simulation vs Experiment

The assignment requires an explicit comparison between predicted and measured behaviour. The comparison must be completed only after a laboratory dataset exists. Use the same setpoint transition and ambient condition for both runs, and tune the identified thermal model using the actual open-loop step test rather than inventing experimental agreement.

Performance indicator	Model prediction	Lab measurement	Difference
Thermal time constant	160 s (assumption)	To be measured	—
Initial to target cooling step	30 → 22 °C	To be logged	—
Settling time	367.3 s	To be measured	—
Cooling undershoot	9.48%	To be measured	—
Final temperature	22.00 °C	To be measured	—
Maximum disturbance deviation	See Figure 5	To be measured	—
Recovery time	Model-based; recalculate	To be measured	—

12.1 Recommended comparison procedure

29. Fit plant time constant and gain to the recorded 40% step.
30. Use the identified parameters and recorded ambient temperature in the MATLAB model.
31. Apply the same reference steps and disturbance timestamps as the physical run.
32. Overlay measured and simulated temperature using the same time axis and units.
33. Calculate quantitative differences and identify physical reasons: thermal contact, hot-side temperature, sensor lag, nonlinear TEC efficiency and load variation.

12.2 Error formula

$$\text{Relative error (\%)} = |\text{Measured} - \text{Simulated}| \,/\, |\text{Simulated}| \times 100\%, \text{ when denominator} \neq 0$$

When comparing temperature values, a difference in °C is often more meaningful than a percentage calculated against an absolute Celsius temperature. For response times, reporting both seconds and percentage difference is appropriate.

12.3 Simulation evidence versus physical experimental results

Figures 11–14 are software simulation outputs. They must not be labelled as laboratory measurements. The provided material does not contain timestamped sensor records, calibrated thermometry, supply current measurements or photographs demonstrating achieved physical temperatures. Add a measurement log for the four reference conditions, then compare each logged profile with the corresponding simulation trace using identical initial and disturbance conditions.

Recommended comparison fields: ambient temperature, heatsink temperature, cooling-face temperature, setpoint, sampled sensor temperature, PWM duty, current, disturbance onset, minimum temperature, settling time and final offset. Discuss sensor contact, fan performance, condensation, Peltier electrical limitations, parameter uncertainty and saturation as possible mismatch sources. Until such data are available, the experiment-versus-simulation agreement must be reported as unverified.

13. Discussion

13.1 Ability to reach and maintain the setpoint

Feedback regulation is the central benefit of the design. The controller continuously observes actual plate temperature and adjusts thermoelectric duty instead of relying on a precomputed constant duty. Under the assumed model, the target is reachable because the effective full-duty temperature reduction is 20 °C while the desired reduction is 8 °C. On a real device, reachability is limited by the Peltier heat-pumping curve and heatsink temperature, and should be confirmed before performance claims are made.

13.2 Interpretation of PID actions

Proportional action provides the prompt correction that responds to plate temperature being too high. Integral action accumulates a sustained error and commands enough additional cooling to balance an extra thermal load. Derivative action evaluates the trend of the sensor signal and moderates rapid

correction. This balance is delicate because DS18B20 quantization and the slow thermal plant make unfiltered differentiation undesirable.

13.3 Why real cooling differs from simulation

A simplified first-order model treats net heat pumping as constant per unit duty. In reality, the TEC current, hot- and cold-side temperatures, supply voltage, internal resistance and thermal conduction all influence net cooling. If the heatsink is small or airflow poor, hot-side temperature increases and cooling effectiveness drops. Contact paste thickness, clamping force, condensation and external heat leaks also change the dynamics. Those effects can explain why a controller that looks excellent in software performs less well on the bench.

13.4 Parameter sensitivity and practical stability

Increasing Kp can make the temperature fall faster, but it may cause excessive duty cycling or too much cooling after the target is crossed. Increasing Ki removes steady-state error more aggressively, but can introduce slow oscillation or windup when cooling authority is insufficient. Increasing Kd can oppose rapid warming or cooling trends, but derivative noise and sensor delay may create command fluctuations. Any tuning assessment should therefore examine both temperature and duty time series.

13.5 Disturbance behavior

A warm object causes an immediate increase in incoming heat. In the nominal energy balance this is represented by positive Qd; without feedback it produces a permanent upward offset. An integral controller can compensate for a moderate load provided the required additional duty does not exceed the output ceiling. For a larger disturbance, the actuator saturates and the plate may not recover until the load is reduced.

13.6 Suitability of PID

PID is appropriate for a classroom-scale, slowly varying and reasonably monotonic cooling plate because it can be implemented at low cost, is easy to interpret and can reject steady thermal disturbances. Its weakness is the nonlinear actuator behavior, asymmetric warming and cooling, and the limitation imposed by hot-side heat rejection. A gain-scheduled controller, feedforward hot-side compensation or more detailed nonlinear control may be justified for demanding operation, but would add calibration and implementation complexity.

13.7 Main engineering lesson

The value of the system is not the use of a PID formula by itself. Successful regulation depends equally on a believable thermal model, a properly constructed heat-flow path, a sensor that represents the controlled variable, output constraints, protection against faults and direct comparison between predicted and observed behavior.

13.8 Engineering discussion of the supplied setpoint plots

A temperature controller can track only references within the reachable operating envelope. For the supplied model, the 15 °C and 20 °C references are feasible; the 5 °C reference clearly is not, and the 10 °C reference loses regulation when disturbed. This result does not establish the minimum temperature of every TEC1-12706 prototype, since achievable temperature depends strongly on heatsink dissipation, Peltier voltage/current, ambient conditions, condensation, plate load and thermal contact.

The closely overlapping PI/PID curves at saturation also explain why derivative action may appear to have no effect: the controller's requested power is clipped to its upper bound. Better tuning may reduce excursions for reachable reference values, but cannot replace additional heat-pumping capacity. The appropriate next improvement is to identify the cooling plant experimentally, confirm the disturbance signal topology and compare the controllers only within their non-saturated operating region.

14. Limitations

- Unidirectional actuation: the system actively cools but cannot actively heat without a reversible driver and appropriate control logic.
- Limited thermoelectric capacity: net heat pumping falls as the hot–cold temperature difference increases.
- Simplified model: constant gain and single plate state do not capture the full coupled cold-side/hot-side thermal dynamics.
- Hot-side dependency: insufficient fan flow or heatsink thermal contact may severely degrade cooling.
- Power losses: PWM, MOSFET dissipation, PSU losses and cable voltage drop affect effective electrical input.
- Sensor limitations: quantization, ±0.5 °C class accuracy, placement and waterproof-probe lag affect feedback.
- Moisture: below-dew-point cold surfaces can collect condensation close to electrical components.
- No experimental record supplied: measured settling time, actual hot-side temperature, component temperature rise and run-to-run variability cannot yet be concluded.

14.1 Restrictions on reported findings

A numerical curve that was generated from a nominal model must be described as simulated. Physical proof requires timestamped recorded readings, photographs of the assembly and a repeatable experimental method. The original hotplate heating results are not interchangeable with cooling results, because actuator sign, thermal coupling and the hot-side boundary condition have all changed.

15. Proposed Improvements

15.1 Better model identification

Carry out duty steps at multiple cooling operating points and identify gain and time constant separately. Include a hot-side temperature state when the forced-air heatsink warms over time. These additions can explain changing cooling strength and guide a gain-scheduled regulator.

15.2 Improved PID implementation

Add bumpless setpoint changes, automated tuning from an open-loop trace, calibrated sensor offset, and a more explicit back-calculation anti-windup strategy. A temperature setpoint ramp can reduce abrupt demand and moisture risk. Long-term disturbances could be compensated with measured ambient and hot-side temperatures.

15.3 Hardware and safety

Use a larger verified heatsink, dedicated fan-current monitoring, separate temperature sensing on the hot side, proper fusing, strain relief, moisture barriers, a protective enclosure and insulated thermal surfaces. A purpose-designed current driver can reduce stress on the TEC compared with poorly filtered, high-ripple direct switching.

15.4 Towards an industrial cooling product

A production device would require calibrated sensors, thermal-runaway fault logic, electrical isolation practices, EMC testing, enclosure ingress protection, power-stage derating, documented environmental limits and repeatable acceptance testing. Thermal and electrical hazard analyses should precede use around biological specimens or valuable electronic hardware.

16. Conclusion

A PID-controlled thermoelectric cooling plate offers a practical way to demonstrate modelling and feedback control for a thermal system. The central difference from a hotplate is that the controller increases cooling demand when measured temperature is above the desired setpoint. This reverses the control sign and changes both the physical assembly and the governing energy balance. The cold face must connect to the controlled plate and the hot face must continuously reject the pumped thermal energy plus module electrical losses.

The report developed a first-order energy-balance model, a second-order sensor-inclusive representation, a state-space description, constrained reverse-action PID control and illustrative numerical comparisons of open-loop and closed-loop response. It also provides a hardware design and a detailed experimental plan appropriate for the IE3034 assignment. These modelling results demonstrate a feasible design approach but cannot establish actual cooling-system performance without measured validation.

The next engineering step is to build the safe cooling assembly, collect open-loop step data, identify physical plant parameters, retune the controller and populate the empty practical-results fields. Only then can the group conclude whether its system satisfies the settling-time, accuracy and disturbance-rejection targets.

17. References

[1] Åström, K. J. and Hägglund, T. Advanced PID Control. ISA, 2006.

[2] Ogata, K. Modern Control Engineering, 5th ed. Pearson, 2010.

[3] Çengel, Y. A. and Ghajar, A. J. Heat and Mass Transfer: Fundamentals & Applications. McGraw-Hill.

[4] Skogestad, S. “Simple analytic rules for model reduction and PID controller tuning.” Journal of Process Control, 2003.

[5] Maxim Integrated, DS18B20 Programmable Resolution 1-Wire Digital Thermometer, datasheet (consult current manufacturer revision).

[6] TEC1-12706 thermoelectric module datasheet; specifications vary by manufacturer and must be matched to the purchased module.

[7] Arduino documentation, analogWrite() and digital input/output reference (consult current reference).

[8] MathWorks documentation: pid, pidtune, feedback, step, lsim and Simulink PID Controller (version-specific syntax should be checked).

[9] IE3034: Control Systems Engineering, “PID-Based Real-World Control System Design and Implementation,” assignment handout, 2026.

Note: Consult the actual component data sheets, MATLAB version and university citation format before submitting. These references establish background; they do not constitute experimental proof of prototype performance.

Appendix A – Arduino Cooling PID Firmware

The following is example firmware for a TEC power-stage test bench using a DS18B20 on D2, PWM on D9, an optional analog hot-side thermistor input and 1-second updates. Hardware safety must be designed independently; verify board, libraries and power driver before use. No physical validation of this sketch is claimed.

```
#include <OneWire.h>
#include <DallasTemperature.h>
#include <math.h>

const byte SENSOR_PIN = 2;
const byte TEC_PWM_PIN = 9;
const byte HOT_SENSOR_PIN = A0;  // optional external monitoring
OneWire wire(SENSOR_PIN);
DallasTemperature sensors(&wire);

const float TS = 1.0f;
const float KP = 0.115f;
const float KI = 0.0015f;
const float KD = 0.55f;
const float MAX_DUTY = 0.85f;
const float MIN_SETPOINT = 18.0f;
const float MAX_SETPOINT = 28.0f;
const float MAX_VALID_PLATE = 80.0f;
const float MIN_VALID_PLATE = -10.0f;
float setpoint = 22.0f;
float integral = 0.0f;
float previousTemperature = NAN;
float filteredSlope = 0.0f;
unsigned long lastSampleMs = 0;
bool fault = false;

void disableCooling() {
  analogWrite(TEC_PWM_PIN, 0);
  integral = 0.0f;
```

```
  filteredSlope = 0.0f;
  previousTemperature = NAN;
}

void setup() {
  pinMode(TEC_PWM_PIN, OUTPUT);
  analogWrite(TEC_PWM_PIN, 0);
  Serial.begin(115200);
  sensors.begin();
  sensors.setResolution(12);
  sensors.setWaitForConversion(false);
  sensors.requestTemperatures();
  lastSampleMs = millis();
  Serial.println("ms,plateC,setpointC,duty,fault");
}

void loop() {
  // Enter a new Celsius target as a line through Serial Monitor.
  if (Serial.available()) {
    String line = Serial.readStringUntil('\n');
    float candidate = line.toFloat();
    if (candidate >= MIN_SETPOINT && candidate <= MAX_SETPOINT) {
      setpoint = candidate;
    }
  }
  unsigned long now = millis();
  if ((unsigned long)(now - lastSampleMs) < 1000UL) return;
  lastSampleMs += 1000UL;

  float plate = sensors.getTempCByIndex(0);
  sensors.requestTemperatures();
  bool valid = isfinite(plate) && plate != DEVICE_DISCONNECTED_C
            && plate >= MIN_VALID_PLATE && plate <= MAX_VALID_PLATE;
  // IMPORTANT: Add validated hot-side/fan interlock for your build.
  fault = !valid;
  if (fault) {
    disableCooling();
    Serial.print(now); Serial.println(",nan,0,0,1");
    return;
  }

  float error = plate - setpoint;   // Reverse-action cooling
  if (!isfinite(previousTemperature)) previousTemperature = plate;
  float rawSlope = (plate - previousTemperature) / TS;
  filteredSlope = 0.8f * filteredSlope + 0.2f * rawSlope;
  previousTemperature = plate;

  float command = KP * error + integral + KD * filteredSlope;
  float duty = constrain(command, 0.0f, MAX_DUTY);
  // Conditional integration at actuator limits (anti-windup)
  if ((command > 0.0f && command < MAX_DUTY) ||
      (command >= MAX_DUTY && error < 0.0f) ||
      (command <= 0.0f && error > 0.0f)) {
```

```
    integral += KI * TS * error;
    integral = constrain(integral, -MAX_DUTY, MAX_DUTY);
  }
  analogWrite(TEC_PWM_PIN, (int)roundf(255.0f * duty));
  Serial.print(now); Serial.print(',');
  Serial.print(plate, 4); Serial.print(',');
  Serial.print(setpoint, 2); Serial.print(',');
  Serial.print(duty, 4); Serial.println(",0");
}
```

Firmware limitation: The hot-side analog pin is reserved but not connected to an implemented calibrated cutoff because its sensor type, conversion and threshold have not been specified. A separate fail-safe hot-side/fan protection circuit should be added and tested before extended operation. The fan should never be PWM-switched off together with the TEC.

Appendix B – MATLAB Modelling and Simulation Script

The following MATLAB script implements the assumed cooling model and a discrete reverse-acting PID. It produces illustrative reference, plate-temperature and duty plots. To reproduce the report with experimental values, replace R, C, Qmax and tauSensor with identified quantities. MATLAB code is provided as an editable reference and has not been executed in MATLAB in this report-generation environment.

```
% IE3034 – PID-controlled Peltier cooling: illustrative model
clear; close all; clc;
Tamb = 30; R = 1.0; Cth = 160;
Qmax = 20; tauSensor = 8;
Kp = 0.115; Ki = 0.0015; Kd = 0.55;
dt = 0.1; Ts = 1.0; Tfinal = 4400;
t = 0:dt:Tfinal;
plate = zeros(size(t)); measured = plate;
duty = plate; ref = plate; loadW = plate;
plate(1)=Tamb; measured(1)=Tamb;
integ=0; derivative=0; prevMeasured=Tamb; u=0;
for k = 2:length(t)
   if t(k) < 1200
      ref(k)=24;
   elseif t(k)<2400
      ref(k)=20;
   else
      ref(k)=24;
   end
   if t(k)>=3000 && t(k)<3700
      loadW(k)=4; % warm-object disturbance in watts
   end
   % One controller update every 1 second
   if mod(k-1,round(Ts/dt))==0
      e=measured(k-1)-ref(k); % COOLING error
      slope=(measured(k-1)-prevMeasured)/Ts;
      derivative=0.8*derivative+0.2*slope;
      raw=Kp*e+integ+Kd*derivative;
      u=min(0.85,max(0,raw));
      if (raw>0 && raw<0.85) || ...
```

```
        (raw>=0.85 && e<0) || (raw<=0 && e>0)
         integ=integ+Ki*Ts*e;
      end
      prevMeasured=measured(k-1);
   end
   duty(k)=u;
   dT=((Tamb-plate(k-1))/R +loadW(k)-Qmax*u)/Cth;
   plate(k)=plate(k-1)+dt*dT;
   measured(k)=measured(k-1)+...
      dt*(plate(k)-measured(k-1))/tauSensor;
end
figure;
subplot(2,1,1);
plot(t/60,plate,'LineWidth',1.5); hold on;
plot(t/60,ref,'--'); grid on;
ylabel('Plate temperature (C)'); legend('Plate','Setpoint');
title('Cooling PID: assumed lumped thermal model');
subplot(2,1,2);
plot(t/60,100*duty,'LineWidth',1.5);
grid on; ylabel('PWM duty (%)'); xlabel('Time (min)');
% Continuous linear cooling plant (negative physical gain)
s=tf('s');
G=-R*Qmax/(R*Cth*s+1);
H=1/(tauSensor*s+1);
disp(G); disp(G*H);
% If comparing in negative-feedback conventions, reverse the
% sign consistently to represent the cooling-specific error.
% Evaluate practical stability with sampling and saturation.
```

Appendix C – Data Recording Sheets and Calculation Guide

C.1 Open-loop step identification table

Time (s)	Ambient (°C)	Plate (°C)	Hot side (°C)	Duty (%)
0	—	—	—	40
30	—	—	—	40
60	—	—	—	40
90	—	—	—	40
120	—	—	—	40
180	—	—	—	40
240	—	—	—	40
300	—	—	—	40
420	—	—	—	40
600	—	—	—	40
900	—	—	—	40

C.2 Closed-loop trial summary

Trial	Setpoint	Initial (°C)	Min (°C)	Settling (s)	Error (°C)	Notes
1	22 °C	—	—	—	—	—
2	22 °C	—	—	—	—	—
3	22 °C	—	—	—	—	—

C.3 Disturbance test summary

Load description	Start (s)	End (s)	Peak deviation (°C)	Recovery (s)
Describe warm object	—	—	—	—

C.4 Calculation example (using assumed model only)

For fixed u = 0.40 and a 30 °C room, a model with K = 20 °C/duty predicts the cooling equilibrium as Tss = 30 − 20 × 0.40 = 22 °C. The 63.2% point of the eight-degree cooling drop is 30 − 0.632 × 8 = 24.944 °C. In the ideal first-order nominal plate model, this occurs after τ = 160 s. In a real sensor trace, lag and changes in hot-side temperature may shift that observed crossing time.

C.5 Evidence checklist

- Save original CSV export rather than editing the raw data.
- Include an actual photograph of the working cold-plate/heatsink assembly.
- Show the real wiring or an exported CAD schematic rather than only a conceptual diagram.
- Capture at least one genuine MATLAB plot generated with the identified plant parameters.
- Record measured fan and hot-side temperatures.
- Check whether observed cooling performance meets the declared targets.

Appendix D – Individual Contributions and Verification Checklist

Member	Existing assigned responsibility	Report verification evidence to supply
M.I.N. Aara	Testing & report	Actual CSV logs, metric calculations, proofread report
M.T.M. Dushan	Hardware	Assembly photos, connections, component ratings
B. Amsawardan	Firmware & PID	Compiled tested code, gain settings, serial capture
M.V.M Fernando	Modelling & simulation	MATLAB project, fitted

parameters, graphs

The allocation above reflects the student details shown in the original submitted hotplate report. Edit the allocation to reflect the group's actual cooling-system contributions before submission.

D.1 Final submission checklist

☐ Lecturer has approved the cooling-system application.

☐ Prototype was demonstrated operating safely.

☐ Cooling face and hot face orientation were verified.

☐ Hot-side heat rejection and fan interlock were tested.

☐ Real test results replace "To be measured" fields.

☐ Simulation outputs were regenerated using identified parameters.

☐ All figure captions correctly distinguish simulated from measured data.

☐ Report is exported to PDF and code/schematic files are included as required.

END OF REPORT